

FedMAQ: Communication-Efficient Federated Learning via Multi-Adaptive Quantization and Knowledge Distillation

BS/MS Computer Science Thesis Update

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We propose a hybrid codesign addressing both system and statistical heterogeneity:

- **Client-Side Compression:** Clients dynamically compress payload bit-widths using multi-adaptive quantization based on memory constraints, dataset sizes, and gradient norms. We draw inspiration from the doubly-adaptive scaling principles in (Hönig et al., 2022).
- **Server-Side Aggregation:** To mitigate quantization noise and non-IID client drift, the server aggregates client logits via proxy-based knowledge distillation. This isolates the computational distillation burden entirely to the server, as demonstrated in (He et al., 2025).

FedMAQ's Multi-Adaptive Quantization

FedMAQ dynamically adjusts each client's transmission bit-width based on three dimensions of awareness:

1. Resource Awareness (Memory Constraint):

Acts as a hard ceiling. It limits the maximum allowable bit-width to prevent Out-of-Memory (OOM) failures on resource-constrained edge devices (He et al., 2025).

2. Data Awareness (Dataset Size):

Clients with larger local datasets possess more representative statistical distributions. FedMAQ allocates higher precision to these clients to preserve their updates (Hönig et al., 2022).

3. State Awareness (Gradient Norm):

Adapts to the local optimization state. High gradient norms indicate large model updates, while low norms suggest proximity to a local minimum (Cui et al., 2026; Liu et al., 2023).

Baselines

Category	Algorithms	Architectural Note
Seminal Controls	FedAvg, FedProx	Standard parameter-averaging without compression.
Pure Quantization	FedPAQ, DAdaQuant	Uplink compression without function-space error mitigation.
Pure Distillation	FedMD, FEDDISTILL+	Sharing logits/soft-labels to mitigate non-IID drift. We benchmark against the FEDDISTILL+ configuration (Jeong et al., 2018).
Hybrid Q+KD	FedKD, CFD	Combines compression and distillation.
Proposed	FedMAQ	Combines multi-adaptive quantization with server-side distillation.

*Note: Algorithms with no public implementations (DynFed, FedDT) were explicitly excluded.

[Cue: Switch to the manuscript]

Next Steps

- Review and optimize existing baseline algorithms replications.
- Refine theoretical framework and methodology chapter narratives.

Thank You!

Questions & Answers